2-Dimensional Toric Code Model

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Introduction

The toric code was first introduced by Alexei Kitaev in 1997, which gets its name from its periodic boundary conditions, giving it the shape of a torus.

Introduction

The toric code was introduced to build the simplest exactly solvable model that exhibits topological order, supports anyons, and stores quantum information nonlocally in a way naturally protected against local errors.

- **Beyond Landau theory**: To have a phase not characterized by local symmetry breaking.
- **Topological order in explicit form**: To exhibit ground-state degeneracy depending on the manifold's topology.
- **Anyon physics**: To realize excitations with nontrivial braiding statistics.
- **Quantum error correction**: To encode qubits globally so that local perturbations are less harmful.
- **Exact solvability**: To have a model simple enough that all these features can be demonstrated analytically.

Introduction

15.2 The toric code

Next we'll discuss a specific CSS code known as the *toric code*, which was discovered by Alexei Kitaev in 1997. In fact, the toric code isn't a single code, but rather it's a family of codes, one for each positive integer starting from 2. These codes possess a few key properties:

- The stabilizer generators have *low weight*, and in particular they all have weight four. In coding theory parlance, the toric code is an example of a quantum low-density parity check code, or *quantum LDPC code* (where *low* means 4 in this case). This is nice because each stabilizer generator measurement doesn't need to involve too many qubits.
- The toric code has *geometric locality*. This means that not only do the stabilizer generators have low weight, but it's also possible to arrange the qubits spatially so that each of the stabilizer generator measurements only involves qubits that are close together. In principle, this makes these measurements easier to implement than if they involved spatially distant qubits.
- Members of the toric code family have increasingly *large distance* and can tolerate a relatively *high error rate*.

Introduction

3 Topological orders in 2d

In 1982, the fractional quantum Hall effect (FQHE) was experimentally discovered by Tsui, Stormer and Gossard [TSG82]. Soon after, it was realized that there are two important features of the FQHE:

1. Tao and Wu [TW84] realized that a fractional quantum Hall state defined on a nontrivial surface has nontrivial ground state degeneracy (GSD). The source of this nontrivial GSD was unclear and caused a lot of confusion, which was often misguided by symmetry-breaking tradition (see for example [And83, NTW85, Tho85]). It was finally clarified by Wen and Niu [Wen90, WN90] that the GSD is robust against arbitrary weak perturbations. Hence the nontrivial GSD is not related to any symmetry and beyond Landau's paradigm.

2. The quasi-particles in a fractional quantum Hall states can carry fractional charges [Lau83] and have fractional statistics [ASW84, Hal84] or even non-abelian statistics [Wu84, Wen91, RM92]. These quasi-particles are also called anyons [Wil82] due to their anyonic statistics. Moreover, the nontrivial GSD is directly related to the fractional statistics of quasi-particles [WN90].

The emergence of anyons and the robust GSD indicate a new kind of order that is beyond Landau's paradigm. The proposed new kind of order was named "topological order" by Wen [Wen90] because its low energy effective theory is a topological quantum field theory (see Remark 2.1.4).

The anyons are special cases of topological defects introduced in Section 2.4 because they are particle-like. In this section we explain in detail that particle-like topological defects in a 2d topological order form a mathematical structure called a unitary modular tensor category.

Definition of the 2d toric code model

- There is a spin-$1/2$ (i.e. a qubit) on each link of the lattice. The local degree of freedom $\mathcal{H}_i$ on each edge i is a two-dimensional Hilbert space $\mathbb{C}^2$. The total Hilbert space is $\mathcal{H}_{\text{tot}} := \otimes_i \mathcal{H}_i = \otimes_i \mathbb{C}^2$. In particular, we impose periodic boundary conditions, hence a torus.
- The Pauli matrices acting on $\mathbb{C}^2$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

satisfying

$$\sigma_i \sigma_j = \delta_{ij} \mathbb{I} + i \varepsilon_{ijk} \sigma_k, \tag{1}$$

where ε_{ijk} is the totally antisymmetric tensor.

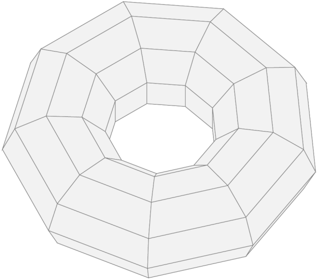


Figure: A 9×9 lattice with periodic boundaries embedded on the surface of a torus.

Definition of the 2d toric code model

For each vertex v and plaquette p we define a vertex operator $A_v = \prod_i \sigma_x^i$ and a plaquette $B_p := \prod_j \sigma_z^j$ acting on adjacent edges. Here $\sigma_x^i = \cdots \otimes 1 \otimes \sigma_x \otimes 1 \otimes \cdots$ is the operator that acts on $\mathcal{H}_i$ as σ_x and acts on other local Hilbert spaces as identities. For example, the operators in following figure are

$$A_v = \sigma_x^1 \sigma_x^2 \sigma_x^3 \sigma_x^4, \quad B_p = \sigma_z^3 \sigma_z^4 \sigma_z^5 \sigma_z^6.$$

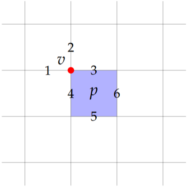


Figure: The 2d toric model.

Definition of the 2d toric code model

$$A_v := \prod_i \sigma_x^i, \quad B_p := \prod_j \sigma_z^j$$

- The Hamiltonian of the toric code model is defined as

$$H := \sum_v (1 - A_v) + \sum_p (1 - B_p), \tag{2}$$

- $[A_v, A'_v] = [B_p, B'_p] = [A_v, B_p] = [A_v, H] = [B_p, H] = 0.$
- $\Rightarrow A_v$ and B_p share common eigenstates. For example,

$$|\psi_\pm\rangle = \frac{1}{\sqrt{2}}(|0000\rangle \pm |1111\rangle), \quad A_v |\psi_\pm\rangle = \pm |\psi_\pm\rangle, \quad B_p |\psi_\pm\rangle = |\psi_\pm\rangle. \tag{3}$$

- The model is exactly solvable.

Definition of the 2d toric code model

$$H := \sum_v (1 - A_v) + \sum_p (1 - B_p)$$

- The total Hilbert space can be decomposed as the direct sum of common eigenspaces of all A_v and B_p operators, whose eigenvalues are ± 1.
- In particular, the ground state subspace has energy 0 and is the common eigenspace of all A_v and B_p operators with eigenvalues $+1$.
- The system is gapped, since the first excited state has energy at least 2, no matter how large the system size is.

Definition of the 2d toric code model

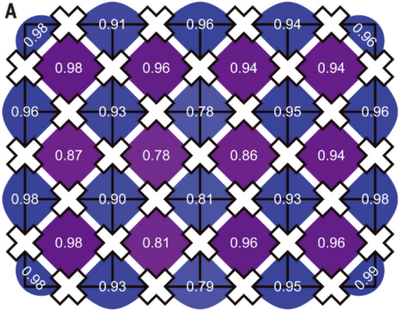


Figure: Experimental implementation of the toric code model by Google.

Ground State Degeneracy

- The toric code model defined on a genus g closed surface. Suppose the lattice has V vertices, E edges and F plaquettes. By Euler's formula, we have

$$V - E + F = 2 - 2g. \tag{4}$$

- The dimension of the total Hilbert space is 2^E, and the ground state subspace is determined by the constraints that all $A_v = 1$ and $B_p = 1$. Note that these constraints are redundant. We have

$$\prod_v A_v = \prod_v \prod_e (\sigma_x)_e^2 = \prod_v \mathbb{I} = 1, \quad \prod_p B_p = \prod_p \prod_e (\sigma_z)_e^2 = \prod_v \mathbb{I} = 1, \tag{5}$$

because the model is defined on a closed surface.

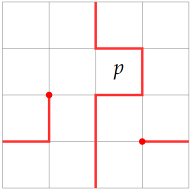
- There are only $(V + F - 2)$ independent constraints, hence the GSD is $2^{E-(V+F-2)} = 2^{2g}$, which is a topological invariant. This is called *topological order*.

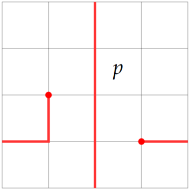
String Configurations

- Choose eigenstates $|\pm\rangle$ of σ_x with eigenvalue ± 1. Consider actions of σ_z:

$$\sigma_z |+\rangle = |-\rangle \,, \quad \sigma_z |-\rangle = |+\rangle \,. \tag{6}$$

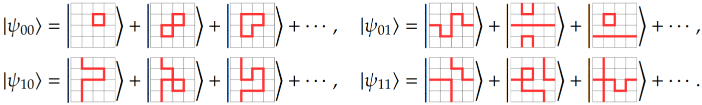
- A string configuration corresponds to a wave function. If there is a red string on an edge, we put a state $|-\rangle$ on this edge, a state $|+\rangle$ otherwise. Each configuration corresponds to a wave function $|+\rangle \otimes |+\rangle \otimes |-\rangle \otimes \cdots$
- The action of the B_p operator locally deforms the string from left to right.





Ground State Degeneracy

- If there is a non-contractible closed string, such a configuration cannot be obtained from the vacuum configuration by B_p operators. Hence, by considering different non-contractible loops on a torus, we get four different (orthogonal) ground state wave functions:



The above states can be characterized by horizontal and vertical winding numbers. The closed strings in them belong to different $\mathbb{Z}_2$-homology classes of a torus.

Topological Excitations of the Toric Code Model

- A *topological excitation* (or a particle-like topological defect) is a subspace of the total Hilbert space that is invariant under the action of local operators.
- Given a state $|\psi\rangle$, the minimal topological excitation containing $|\psi\rangle$ is

$$\{A\,|\psi\rangle \mid A \text{ is a local operator}\},$$

  called the topological excitation *generated* by $|\psi\rangle$. In particular, the topological excitation generated by the ground state subspace is called the *trivial topological excitation*, denoted by $\mathbb{I}$.
- A *nontrivial* topological excitation is usually defined to be an excitation which cannot be created from or annihilated to the ground state by local operators.

Topological Excitations of the Toric Code Model

Given a vertex v_0, there is a state $|\psi_{v_0}\rangle$ satisfying the following property (?):

$$\begin{cases} A_{v_0}|\psi_{v_0}\rangle = -|\psi_{v_0}\rangle, \\ A_v|\psi_{v_0}\rangle = |\psi_{v_0}\rangle \text{ for all vertices } v \neq v_0, \\ B_p|\psi_{v_0}\rangle = |\psi_{v_0}\rangle \text{ for all plaquettes } p. \end{cases}$$

In other words, $|\psi_{v_0}\rangle$ is an excitation located at v_0.

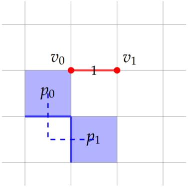


Figure: Topological excitations of toric code.

Topological Excitations of the Toric Code Model

Then we try to apply a local operator σ_z^1 on $|\psi_{v_0}\rangle$. Since σ_z^1 commutes with all plaquette operators and vertex operators except A_{v_0} and A_{v_1}, we still have

$$B_p \cdot \sigma_z^1 |\psi_{v_0}\rangle = \sigma_z^1 |\psi_{v_0}\rangle$$

$$A_v \cdot \sigma_z^1 |\psi_{v_0}\rangle = \sigma_z^1 |\psi_{v_0}\rangle$$

for all plaquettes p and vertices $v \neq v_0, v_1$.

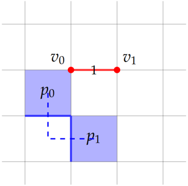


Figure: Topological excitations of toric code.

Topological Excitations of the Toric Code Model

But σ_z^1 anti-commutes with A_{v_0} and A_{v_1}, so we have

$$A_{v_0} \cdot \sigma_z^1 |\psi_{v_0}\rangle = -\sigma_z^1 A_{v_0} |\psi_{v_0}\rangle = \sigma_z^1 |\psi_{v_0}\rangle,$$
$$A_{v_1} \cdot \sigma_z^1 |\psi_{v_0}\rangle = -\sigma_z^1 A_{v_1} |\psi_{v_0}\rangle = -\sigma_z^1 |\psi_{v_0}\rangle.$$

Therefore, $\sigma_z^1 |\psi_{v_0}\rangle$ is an excitation located at v_1.

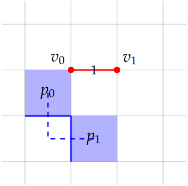


Figure: Topological excitations of toric code.

Topological Excitations of the Toric Code Model.

Local operators cannot annihilate such an excitation $|\psi_{v_0}\rangle$. So $|\psi_{v_0}\rangle$ generates a nontrivial topological excitation (a quasiparticle), denoted by e. All $|\psi_{v_0}\rangle$'s generate the same topological excitation e, because they can be connected by a string of σ_z operators.

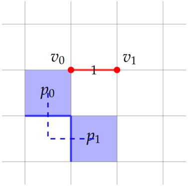


Figure: Topological excitations of toric code.

Topological Excitations of the Toric Code Model

Similarly, for each plaquette p_0 there is a state $|\psi_{p_0}\rangle$ satisfying the following property:

$$\begin{cases} B_{p_0}|\psi_{p_0}\rangle = -|\psi_{p_0}\rangle, \\ B_p|\psi_{p_0}\rangle = |\psi_{p_0}\rangle \text{ for all plaquettes } p \neq p_0, \\ A_v|\psi_{p_0}\rangle = |\psi_{p_0}\rangle \text{ for all vertices } v. \end{cases}$$

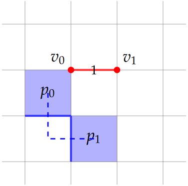


Figure: Topological excitations of toric code.

Topological Excitations of the Toric Code Model

We have

$$B_{p_0}\sigma_x^1|\psi_{p_0}\rangle = -\sigma_x^1 B_{p_0}|\psi_{p_0}\rangle = \sigma_x^1|\psi_{p_0}\rangle,$$
$$B_{p_1}\sigma_x^1|\psi_{p_0}\rangle = -\sigma_x^1 B_{p_1}|\psi_{p_0}\rangle = -\sigma_x^1|\psi_{p_0}\rangle.$$

Then all states $|\psi_{p_0}\rangle$ generate the same topological excitation, denoted by m.

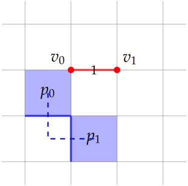


Figure: Topological excitations of toric code.

Topological Excitations of the Toric Code Model

- For every vertex v_0 and plaquette p_0 there is a state $|\psi_{v_0,p_0}\rangle$ satisfying the following property:

$$\begin{cases} A_{v_0}|\psi_{v_0,p_0}\rangle = -|\psi_{v_0,p_0}\rangle, \\ A_v|\psi_{v_0,p_0}\rangle = |\psi_{v_0,p_0}\rangle \text{ for all vertices } v \neq v_0, \\ B_{p_0}|\psi_{v_0,p_0}\rangle = -|\psi_{v_0,p_0}\rangle, \\ B_p|\psi_{v_0,p_0}\rangle = |\psi_{v_0,p_0}\rangle \text{ for all plaquettes } p \neq p_0. \end{cases}$$

  Then all states $|\psi_{v_0,p_0}\rangle$ generate the same topological excitation, denoted by f.

- We have found 4 different topological excitations: $\mathbb{I}$, e, m, f.
- Intuitively, if an e particle and an m particle are adjacent, they can be viewed as a single topological excitation f.

Fusion Rules

- The definition of the excitation f:

$$\begin{cases} A_{v_0}|\psi_{v_0,p_0}\rangle = -|\psi_{v_0,p_0}\rangle, \\ A_v|\psi_{v_0,p_0}\rangle = |\psi_{v_0,p_0}\rangle \text{ for all vertices } v \neq v_0, \\ B_{p_0}|\psi_{v_0,p_0}\rangle = -|\psi_{v_0,p_0}\rangle, \\ B_p|\psi_{v_0,p_0}\rangle = |\psi_{v_0,p_0}\rangle \text{ for all plaquettes } p \neq p_0. \end{cases}$$

- Intuitively, if an e particle and an m particle are adjacent, they can be viewed as a single topological excitation f. This indeed gives the *fusion rule* (algebra):

$$e \times m = m \times e = f. \tag{7}$$

Similarly,

$$e \times e = m \times m = f \times f = \mathbb{I}, \quad f \times e = m, \quad f \times m = e. \tag{8}$$

Fusion Rules

Fusion rules, describing the behavior when two particles are pulling closer, can be equivalently realized by the operator product expansion (OPE):

$$\hat{\phi}_a(x)\hat{\phi}_b(y) = C_{abc}(x, y, z)\hat{\phi}_c(z), \tag{9}$$

where the Einstein summation convention is applied.

Braiding Statistics

- Recall: A state $|\psi_m\rangle$ with an m at a plaquette p_0 satisfies $B_{p_0} = -1, B_p = 1 (p \neq p_0), A_v = 1.$
- Let $|\psi_0\rangle$ be the ground state. Define a *string operator*

$$W_m(\gamma) := \prod_{i \in \gamma} \sigma_i^x. \tag{10}$$

Then

$$|\psi_m\rangle = W_m(\gamma) |\psi_0\rangle . \tag{11}$$

- Let C be a closed loop on the lattice. Define

$$L_e(C) = \prod_{j \in C} \sigma_j^z. \tag{12}$$

$L_e(C)$ creates no excitations itself.

Braiding Statistics

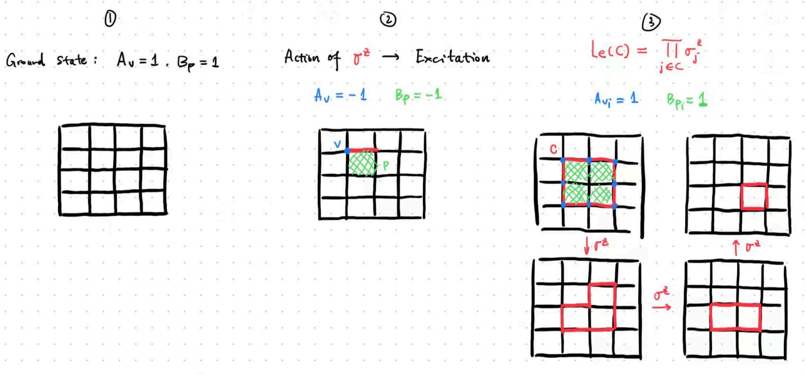


Figure: Illustration of the operator $L_e(C)$. If the path C forms a loop, then $L_e(C)$ creates no excitation. Otherwise not.

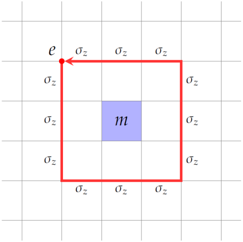
Braiding Statistics

- Denote the intersection edge as k, then

$$
\begin{aligned}
L_e(C)\,W_m(\gamma)\,|\psi_0\rangle &= \left(\sigma_k^z \prod_{j\neq k}\sigma_j^z\right)\left(\sigma_k^x \prod_{i\neq k}\sigma_i^x\right)|\psi_0\rangle \\
&= \sigma_k^z\sigma_k^x \prod_{j\neq k}\sigma_i^z \prod_{i\neq k}\sigma_i^x\,|\psi_0\rangle \\
&= -\sigma_k^x\sigma_k^z \prod_{i\neq k}\sigma_i^x \prod_{j\neq k}\sigma_i^z\,|\psi_0\rangle \\
&= -\,W_m(\gamma)L_e(C)\,|\psi_0\rangle\,.
\end{aligned}
\tag{13}
$$

Braiding Statistics

- $L_e(C)$ represents: create an e, move it around the loop C, then annihilate it. Physically, this means: an e particle winding once around an m particle introduces a phase -1.



- By topological equivalence, they exchange twice, each swap produces a phase $i = e^{i\pi/2}$. Hence the name *anyon*.
- The same for an m going around an e.

Summary

- In the 2d toric code model, qubits live on links of a square lattice. The model is gapped and exactly solvable.
- Ground state satisfies $A_v = B_p = +1$ for all v, p. On a surface of genus g, the ground state degeneracy is 2^{2g}. Wave functions can be characterized by horizontal and vertical winding numbers.
- Topological excitations $\mathbb{I}, e, m, f$ exhibit nontrivial statistics, distinct from bosons and fermions. Fusion rules describe what remains if they are brought together locally.

Thanks for your attention!

Backup

As shown in figure 2, the commutator between A_v and B_p operators can be straightforwardly calculated:

$$
\begin{aligned}
[A_v, B_p] &= [\sigma_x^1\sigma_x^2\sigma_x^3\sigma_x^4, \sigma_z^3\sigma_z^4\sigma_z^5\sigma_z^6] \\
&= \sigma_x^1\sigma_x^2[\sigma_x^3\sigma_x^4, \sigma_z^3\sigma_z^4\sigma_z^5\sigma_z^6] + [\sigma_x^1\sigma_x^2, \sigma_z^3\sigma_z^4\sigma_z^5\sigma_z^6]\sigma_x^3\sigma_x^4 \\
&= \sigma_x^1\sigma_x^2\left(\sigma_z^3\sigma_z^4[\sigma_x^3\sigma_x^4, \sigma_z^5\sigma_z^6] + [\sigma_x^3\sigma_x^4, \sigma_z^3\sigma_z^4]\sigma_z^5\sigma_z^6\right) \\
&= \sigma_x^1\sigma_x^2\left(\sigma_x^3\sigma_z^3[\sigma_x^4, \sigma_z^4] + [\sigma_x^3, \sigma_z^3]\sigma_x^4\sigma_z^4\right)\sigma_z^5\sigma_z^6 \\
&= \sigma_x^1\sigma_x^2\left(-i\sigma_x^3\sigma_z^3\sigma_y^4 - i\sigma_y^3\sigma_z^4\sigma_z^4\right)\sigma_z^5\sigma_z^6 \\
&= -i\sigma_x^1\sigma_x^2\left(\varepsilon_{xzy}\sigma_y^3\cdot\sigma_y^4 + \varepsilon_{zxy}\sigma_y^3\cdot\sigma_y^4\right)\sigma_z^5\sigma_z^6 \\
&= 0.
\end{aligned}
\tag{14}
$$